Assessment Across the Lifespan: Integumentary & Lymphatic Systems

NURA 330 Health Assessment | Module 4 Lecture Study Guide

Exam 2 content · reviewed for the final

Lecture roadmap: Case: Ms. P → A&P: skin layers and lymphatic system → Subjective assessment (OLDCARTS for skin, ABCDE) → Psychosocial and SDOH → Skin assessment in people of color → Pressure injury stages → Braden Scale → Medical device-related pressure injuries → Tunneling vs. undermining → Red flags → Expected vs. unexpected findings and lymph nodes → Terminology and nursing diagnoses → Lifespan (fontanelles, older adults) → Documentation → Care plan for Ms. P

Exam 2 study guide topics: growth and development across the lifespan; skin and lymph A&P; subjective history; psychosocial impact and SDOH; **pressure injury stages**; **Braden Scale**; **MDRPI**; **tunneling vs. undermining**; red flags; expected vs. unexpected findings.

How to read this guide: Everything without a tag comes from the lecture slides or the slide notes. Anything else is tagged: STUDY GUIDE emphasized on the exam study guide but not spelled out on the slides; ADDED information not in the slides; CLARIFIED a slide point explained further; CORRECTED an error or inconsistency in the slides. Every tag is also listed in the notes at the end.

Case: Ms. P, 78. Admitted with a severe UTI, sepsis, and fever of 101.4 °F. **Bedbound, diaphoretic, lethargic, incontinent.** On turning, her **sacrum is red and stays red** after pressure is relieved. Braden score **14 (moderate risk)**: limited mobility, moisture, friction/shear. Challenge: is this a **Stage 1 pressure injury** or **moisture-associated skin damage (MASD)**?

Anatomy and Physiology

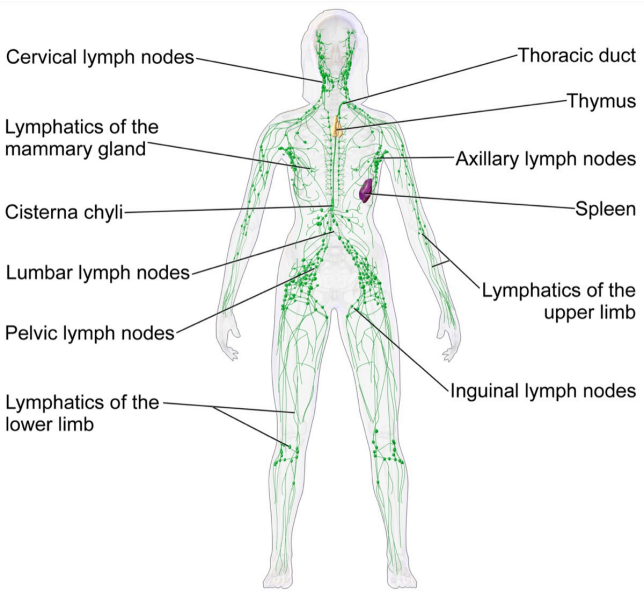
Integumentary System

Layer	Key features
Epidermis	**Avascular** outer protective barrier, mostly keratinocytes; **replaced every 3–4 weeks**

Layer	Key features
Dermis	**Vascular** dense connective tissue: collagen, elastic fibers, nerve endings, hair follicles, sweat and oil (sebaceous) glands
Hypodermis (subcutaneous)	Fat: **insulation, energy storage, cushioning** over bony prominences

Lymphatic System

- **Lymph vessels** collect excess interstitial (tissue) fluid.
- **Regional node chains: cervical, axillary, epitrochlear** (inner elbow), and **inguinal**.
- **Functions:** fluid balance, immune filtration, pathogen defense.



Lymphatic system and major node groups (lecture slide)

Subjective Assessment

OLDCARTS for Skin and Lymph Symptoms

Onset	When did the lesion, rash, or swelling start?
Location	Localized, regional, or generalized?
Duration	Persistent, constant, or intermittent?
Character	Itching (pruritus), burning, aching, sharp
Aggravating	Heat, sweating, soaps, clothing?

Relieving	Cool compresses, elevation, topicals?
Timing	Time of day, season, menstrual cycle?
Severity	0–10; effect on daily activities

Also ask about: prior skin history (eczema, psoriasis, skin cancer, slow wound healing); color changes (hyperpigmentation, cyanosis, jaundice, pallor); **changes in moles (ABCDE)**; medications and topicals (including corticosteroids); sun exposure, tanning beds, sun protection, and occupational chemicals; lymphatic symptoms (swollen glands, tenderness, **night sweats, unexplained weight loss**).

ADDED **ABCDE for moles (melanoma warning signs):** **A**symmetry; **B**order irregular; **C**olor varies within the lesion; **D**iameter larger than 6 mm (a pencil eraser); **E**volving (changing size, shape, color, or new symptoms). The slide names ABCDE but does not spell it out.

Psychosocial and SDOH

- **Body image:** visible lesions, acne, scars, or chronic ulcers affect self-esteem.
- **Emotional distress:** chronic itching or psoriasis flares cause sleep loss, isolation, anxiety.
- **Stress:** worsens inflammatory skin disease (eczema, psoriasis, hives) through cortisol release.
- **Coping:** support systems, coping strategies, willingness to take part in wound care.
- **Housing and hygiene access:** clean running water, sanitation, shelter for wound care.
- **Financial barriers:** ability to buy dressings, barrier creams, antibiotics.
- **Environmental exposures:** sun, pesticides, industrial chemicals.
- **Abuse and trauma screening:** inconsistent injury patterns, e.g., **multiple bruises in different stages of healing**; follow trauma-informed reporting protocols.
- **Gender-affirming care:** sexual orientation and gender identity are different; use the patient's pronouns and avoid assumptions during the exam.

Skin Assessment in People of Color

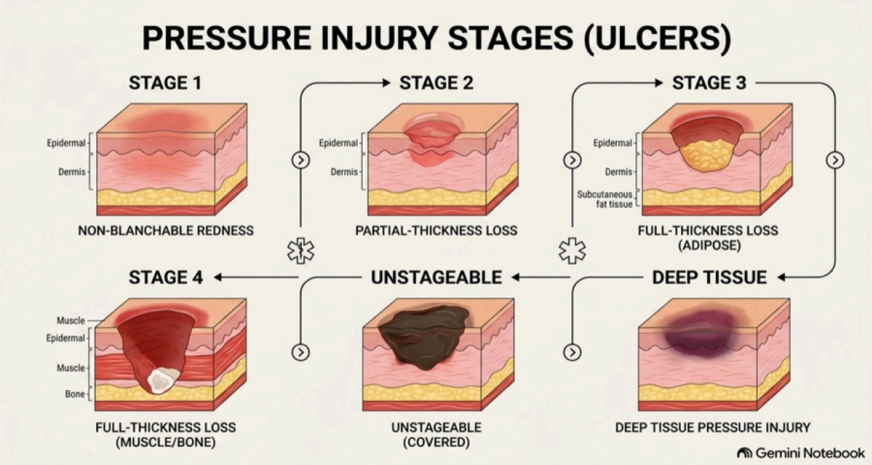
Finding	How it appears in darker skin / where to look
Best sites overall	**Sclera, conjunctiva, oral mucosa, palms, soles** for cyanosis, pallor, and jaundice. Use **palpation** (warmth, edema, induration) rather than relying on redness

Finding	How it appears in darker skin / where to look
Stage 1 pressure injury	Does **not** look bright red. Appears **dark purple, eggplant, or violaceous**, with **local warmth** and **boggy or firm** tissue
Cyanosis	Dull **gray or bluish** tones in oral mucosa, soft palate, conjunctiva
Pallor	**Ash-gray or yellowish-brown** skin; conjunctiva loses its normal pink
Jaundice	Best seen in the **hard palate** and **sclera** (do not confuse with normal carotene deposits at the outer sclera)
Baseline	**Always ask the patient or family what is normal** for their skin

Pressure Injuries

Stages

Stage	What you see
Stage 1	Intact skin with **non-blanchable erythema** (or dark discoloration in darker skin)
Stage 2	**Partial-thickness** skin loss with **exposed dermis**; may be an intact or ruptured **blister**
Stage 3	**Full-thickness** skin loss; **fat (adipose) visible**
Stage 4	Full-thickness loss with **exposed bone, muscle**, or tendon
Unstageable ADDED	Full-thickness loss where the base is **covered by slough or eschar**, so depth cannot be seen (shown in the slide's diagram)
Deep tissue pressure injury ADDED	Intact or non-intact skin with persistent **deep red, maroon, or purple** discoloration or a blood-filled blister, from damage to deeper tissue (shown in the slide's diagram)



Pressure injury stages (lecture slide)

ADDED **Blanchable vs. non-blanchable:** Press on the red area for a few seconds. If it turns white and the color returns, it is **blanchable** (pressure is being relieved; not yet an injury). If it **stays red**, it is **non-blanchable**, which is a Stage 1 pressure injury. Ms. P's sacrum "remains persistently red after pressure relief," so it fits Stage 1. **MASD**, by contrast, is caused by moisture (urine, sweat), tends to be diffuse and shiny over skin folds or the perineum rather than over a bony prominence, and may have irregular edges.

Braden Scale

Six subscales	**Sensory perception, Moisture, Activity, Mobility, Nutrition** (each 1–4 points) and **Friction/Shear** (1–3 points)
Total	6 (worst) to 23 (best). **Lower score = higher risk**
Risk levels	**15–18:** mild risk | **13–14:** moderate | **10–12:** high | **9 or below:** very high (very severe). 19–23 = no risk (slide image)
Critical thinking	**Do not rely on the total alone.** A low subscale (e.g., Moisture = 1) needs targeted intervention right away



Braden Scale (lecture slide)

Medical Device-Related Pressure Injuries (MDRPI)

- **Definition:** pressure injuries from devices used for diagnosis or treatment; the injury **takes the shape of the device**.
- **High-risk devices:** CPAP/BiPAP masks, nasal cannulas, ET tubes, trach ties, Foley catheters, pulse oximeter probes, rigid splints.
- **Key rule: the Braden Scale does NOT account for device pressure.** Remove or reposition devices to inspect the skin **at least twice daily**.
- **Prevention:** thin hydrocolloid under masks; foam ear protectors on O_2 tubing; **rotate pulse ox probe sites every 4 hours**; secure Foleys without tension.

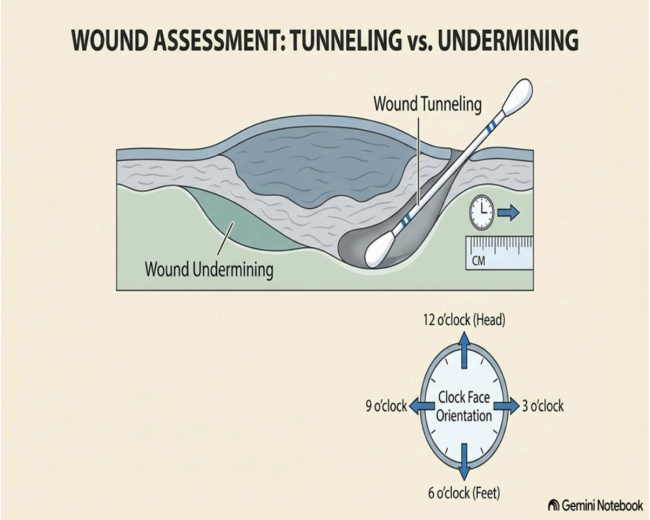
Device	Where injury happens	Prevention
Nasal cannula / O_2 tubing	Top of the ears, cheeks	Foam tubing pads; check skin every 8 hours
CPAP / BiPAP mask	Bridge of nose, forehead	Hydrocolloid barrier; check mask fit
ET / trach tube	Lips, corners of mouth, neck	Reposition ET tube daily; foam ties
Foley / rigid lines	Inner thigh, meatus	Securement device; avoid tension

CORRECTED The slides give two inspection frequencies (at least twice daily on one slide; every 8 hours for nasal cannula on the other). Follow your agency's policy; see note 1.

Wound Tunneling vs. Undermining

	Tunneling	Undermining
What it is	A **narrow channel** extending in any direction from the wound into deeper tissue (fascia, muscle)	Tissue destruction **under intact skin along the wound edges** (a wide "shelf" or "lip")

- **Clock-face orientation: 12 o'clock = toward the patient's head**; 6 o'clock = toward the feet.
- **Measuring:** gently insert a **sterile cotton-tipped applicator** until you feel resistance → grasp the stick at the wound edge with a gloved finger → remove → measure against a **metric ruler in cm**.
- **Charting example:** "Tunneling noted at 2 o'clock position extending 1.8 cm deep."



Tunneling vs. undermining and clock-face orientation (lecture slide)

Red Flags: Prioritize and Escalate

Finding	Why / action
Central cyanosis	Arterial hypoxia: **immediate airway/oxygen intervention**; call the provider now
Petechiae & purpura	Possible bleeding disorder, thrombocytopenia, or sepsis/meningococcemia
Rapidly spreading cellulitis	**Mark the red border with a skin marker**; obtain blood and wound cultures; escalate for IV antibiotics

Finding	Why / action
Acute jaundice	Possible liver failure: check LFTs and bilirubin; watch for hepatic encephalopathy (confusion)
Hard, fixed lymph nodes	>1 cm, firm, non-tender, fixed: possible lymphoma or metastatic cancer; refer for urgent biopsy

Expected vs. Unexpected Findings

Domain	Expected	Unexpected
Color & perfusion	Even tone for genetic background; warm, dry; pink mucous membranes; cap refill <2 sec	Central cyanosis (blue lips), jaundice (yellow sclera), pallor, persistent non-blanchable redness
Turgor & elasticity	**Prompt recoil (<2 sec)** when pinched over the **clavicle or sternum**	**Tenting** (>2 sec): dehydration; **maceration** from chronic moisture
Lymph nodes	Not palpable, or if palpable: **<1 cm, soft, mobile, smooth, non-tender**	**Infection:** enlarged, **tender**, firm, **mobile**. **Malignancy: hard, fixed, non-tender**, >1 cm
Lesions & integrity	Intact, no breakdown, tears, or abnormal lesions	Stage 1–4 injuries, petechiae, spreading cellulitis, purulent drainage, tunneling

Lymph node memory aid: **Tender + mobile = infection** (the body fighting something). **Hard + fixed + painless = worry about cancer.**

Terminology and Nursing Diagnoses

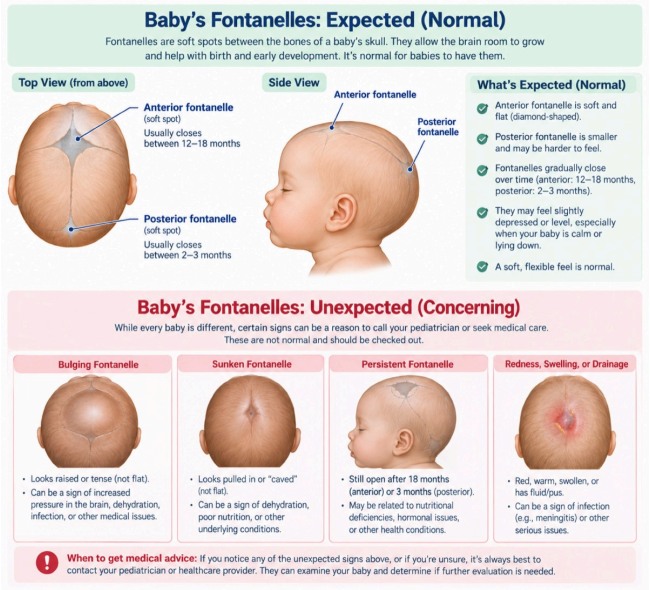
Term	Meaning
Erythema	Redness from capillary congestion
Ecchymosis	Purple discoloration from blood leaking into tissue (bruise)
Diaphoresis	Profuse sweating (fever, shock)

Term	Meaning
Maceration	Soft, white, broken-down skin from chronic moisture
Exudate	Wound drainage: serous (clear), sanguineous (bloody), purulent (pus)
Induration	Firm, hardened skin from inflammation
Eschar	Thick black/brown dead tissue

NANDA diagnoses: Impaired Skin Integrity (epidermis/dermis); **Risk for Impaired Skin Integrity** (fever, diaphoresis, thin skin, shear); **Impaired Tissue Integrity** (subcutaneous tissue, muscle, bone: Stage 3–4); **Risk for Infection** (broken barrier, tunneling wounds, lines); **Acute Pain**.

Lifespan Considerations

Age group	Assessment approach and findings
Infants & toddlers	Observe first, non-invasively, while the child is calm on the caregiver's lap. Palpate fontanelles: **anterior closes 12–18 months**; **posterior closes 1–2 months** CORRECTED (see note 2). **Sunken = dehydration; bulging = increased ICP**
Children & adolescents	**Speak to the child directly** to build trust; provide privacy and respect modesty
Older adults	Thinner epidermis, less subcutaneous fat, less elasticity; **senile purpura** (easy bruising) and **lentigines** (age spots). **High risk for skin tears and sacral breakdown**; handle skin gently



Expected vs. concerning fontanelle findings (lecture slide)

STUDY GUIDE ADDED **Age-appropriate approach across the lifespan** (the study guide's lifespan objectives go beyond the one lifespan slide):

- **Infants:** do quiet assessments first (heart, lungs) while calm; save invasive or upsetting parts (ears, mouth) for last.
- **Toddlers/preschoolers:** let them touch the equipment; use play; keep the caregiver close.
- **School-age:** explain simply what you are doing and why; they value privacy and like to help.
- **Adolescents:** interview privately for part of the visit; confidentiality matters.
- **Older adults:** allow extra time; make sure glasses and hearing aids are in; compare findings with *their* baseline; watch for atypical presentations (confusion as a sign of infection).

Documentation

Do not chart	Why
"Skin looks fine overall."	Vague, not objective
"Small pressure sore on bottom."	No measurements or anatomical landmarks

Do not chart	Why
"Wound looks better today."	No objective comparison to baseline
"Skin is red."	Does not say blanchable or non-blanchable
"Slight drainage from wound."	Missing color, amount, odor, type

Gold-standard note (from the slides): "Skin intact, warm, and dry throughout. Right sacral region presents with a 2.5 cm x 3.0 cm area of non-blanchable erythema without skin breakdown or open erosion. Surrounding tissue is intact with localized warmth on palpation. Braden Scale score calculated at 14. Patient turned to left lateral position with 30-degree wedge support. Heels floated on pillows off mattress bed. Dimethicone barrier cream applied to perineum following hygiene care."

Care Plan: Ms. P

Risk analysis (Braden 14)	Targeted interventions
Moisture = 2: diaphoresis from fever + incontinence	**Dimethicone barrier cream** after each incontinence episode; change linens promptly
Mobility = 2: very limited movement from sepsis lethargy	**Turn every 2 hours** using a **30-degree lateral tilt** to relieve the sacrum; **float heels** on pillows under the lower legs
Friction/shear = 1: slides down when HOB is above 30°	**Keep HOB at 30° or lower**; use a lift sheet to reposition (no dragging)
Goal: keep the sacral redness from progressing to an open Stage 2	**Wound care nurse consult** and **high-protein nutrition consult**

Quick Review

Topic	Key points
Skin layers	Epidermis (avascular, 3–4 week turnover); dermis (vascular); hypodermis (fat)
Node chains	Cervical, axillary, epitrochlear, inguinal
Darker skin	Check sclera, conjunctiva, mucosa, palms, soles; Stage 1 = purple/eggplant + warmth/bogginess
Stages	1 non-blanchable; 2 partial thickness/dermis; 3 fat; 4 bone/muscle; unstageable = covered; DTPI = purple/maroon
Braden	6 subscales; 6–23; lower = worse; ≤18 at risk; check subscales
MDRPI	Braden misses it; inspect under devices at least twice daily
Tunneling vs. undermining	Channel vs. shelf under edge; 12 o'clock = head; sterile applicator, cm
Turgor	Over clavicle/sternum; tenting >2 sec = dehydration
Nodes	Tender + mobile = infection; hard + fixed + non-tender >1 cm = malignancy
Fontanelles	Anterior closes 12–18 mo; posterior early; sunken = dehydration; bulging = ↑ICP

Self-Check ADDED

1. A Black patient's sacrum is not red but feels warm and boggy with a purple tone. What does this suggest? *Answer: A Stage 1 pressure injury.*
2. A wound has a shallow open area with pink dermis and no visible fat. Stage? *Answer: Stage 2.*
3. Braden score of 11. Risk level? *Answer: High risk.*
4. A patient on BiPAP has a Braden of 21. Is the patient at risk for a pressure injury? *Answer: Yes, from the device; the Braden Scale does not account for device pressure.*
5. A neck node is 2 cm, hard, fixed, and painless. Concern? *Answer: Possible malignancy; report for further evaluation.*
6. An infant's anterior fontanelle is sunken. Most likely cause? *Answer: Dehydration.*

Notes on Sources, Additions & Clarifications

Corrections and inconsistencies in the slides:

1. **Device skin checks.** One MDRPI slide says to inspect under devices "at least twice daily"; the device diagram says to check skin under a nasal cannula "q8h." These are not contradictory (q8h is more frequent than twice daily), but know that the stated **minimum is twice daily** and follow agency policy.
2. **Posterior fontanelle closure.** The slide text says "1–2 months," while the slide's fontanelle diagram says "2–3 months." Most references give about 2 months. Ask your instructor which number they want; the anterior fontanelle (12–18 months) is consistent.

From the Exam 2 study guide:

- The topic list in the blueprint box.
- The study guide's lifespan objectives (age-appropriate techniques and communication), which go beyond the single lifespan slide; covered in the age-appropriate approach box.

Added information (not in the slides):

- ABCDE spelled out.
- Unstageable and deep tissue pressure injury definitions (both appear in the slide's staging diagram but not in its text).
- How to check blanching, and how MASD differs from a Stage 1 pressure injury (the case asks you to tell them apart, but the slides do not explain how).
- Braden total range 6–23 (from the slide image, not its text) and the lymph node memory aid.
- The age-by-age approach box and the self-check questions.

Not included: one slide note ("Nail beds may show central slowing") was unclear and left out. Some slide notes in this deck were copied from the respiratory lecture (e.g., about James and lung auscultation) and do not apply here.